

Team Testers  
Final Project Team Reflective Journal  
ITAI 2373  
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## **NewsBot Intelligence System 2.0 Team Reflective Journal**

### **Introduction**

This journal reflects on the Team Testers review process for the NewsBot 2.0 final project. Abraham Barreto and I did not build one shared notebook or combine our code into one system. We built our own individual NewsBot projects and then reviewed each other's systems as peer testers. That setup matched how we actually worked because each of us stayed responsible for an individual submission. The review process still helped a lot because another person could test the project like a real user instead of reading it with the same assumptions as the person who built it.

### **Team Collaboration Analysis**

Team Testers gave us a simple but useful way to collaborate. I tested Abraham's project by checking the notebook flow, trying sample inputs, and looking at whether the outputs made sense. Abraham tested my NewsBot 2.0 notebook and dashboard in the same way, especially checking confusing labels, confidence wording, and places where the explanation needed clearer language. We did not split one system into coding tasks because our projects stayed individual. Instead, we shared the review work, which helped both of us improve quality without pretending we had one combined project.

### **Technical Integration Challenges**

The hardest part of the final project came from connecting many NLP tools into one smooth system. Classification, topic modeling, sentiment analysis, entity extraction, summarization, semantic search, multilingual processing, conversation, and the web interface all had to work together. I had to make each section use the right text field and return output that the next section could use. Abraham's testing helped me see whether the final system made sense to someone who did not already know my variable names or notebook structure. That feedback mattered because a notebook can run without errors and still confuse a real user.

### **Business Value Assessment**

The final NewsBot has business value because it helps users review news articles faster. A communications team could use a similar system to watch company mentions, article tone, and important organizations. A business analyst could use the search and summary features to find related coverage without manually reading every article first. The dashboard also makes the project easier to explain because users can paste an article and quickly see the category, sentiment, summary, entities, and related articles. The project still needs human judgment, but it gives users a strong first-pass review.

### **Individual Contributions**

I built and documented my NewsBot 2.0 system from dataset loading through the final dashboard test. My work included dataset validation, preprocessing, enhanced classification,

topic modeling, sentiment tracking, entity relationship mapping, summarization, semantic search, multilingual processing, conversational interface, testing, evaluation, the Gradio frontend, and the bias/framing extension. Abraham helped my project by reviewing and testing the system from a user's point of view. I helped his project in the same way by checking flow, clarity, and usability. We did not trade ownership of code, but we both improved our final projects through review.

### **Lessons Learned**

This project showed me that integration matters as much as individual techniques. A classifier can score well, but the project becomes more useful when the system also explains the result, summarizes the article, finds entities, and gives the user a clean interface. Comments and labels also mattered because another person needed to understand the notebook without sitting next to me. The Team Testers process made this clear because peer testing exposed usability questions faster than solo testing. The experience felt closer to a real development process where someone else reviews the product before final delivery.

### **Future Enhancements**

In the future, I would improve the project with newer articles, real publication dates, and stronger multilingual support. The final notebook uses generated analysis months because the BBC dataset does not provide verified publication dates, so real dates would make the trend analysis more meaningful. Stronger open-source summarization and embeddings could also improve the system if the runtime can handle them without paid APIs. Another improvement would involve saving trained models and organizing the code into separate Python modules. Those changes would make the project look even more like a production system.

### **Professional Development Impact**

The final project helped us see how classroom NLP ideas can turn into a complete working tool. We practiced loading and validating data, cleaning text, training models, reading evaluation results, creating summaries, extracting entities, and building an interface. We also had to explain technical choices in a way that a user or professor could understand. The Team Testers review helped us practice giving and receiving feedback based on how someone else experienced the system. That matters because real AI projects need both technical quality and clear communication.

### **Conclusion**

Overall, the Team Testers process improved the final NewsBot 2.0 project and the way I explain it. A strong NLP system needs more than code that runs, it needs clear setup, useful output, honest limitations, and testing from another person's perspective. Abraham and I kept our projects individual, but peer review gave the work meaningful collaboration and helped both systems become easier to use. The project also created a stronger portfolio example because it shows the full pipeline from raw article text to dashboard output. If I keep improving it, I want to make it more current, more explainable, and more useful for real news monitoring.